

Institute of Actuaries of India

Subject SA2 – Life Insurance

May 2015 Examination

INDICATIVE SOLUTIONS

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**(i)**

IRDA (Issuance of Capital by Life Insurance Companies) Regulations, 2011 (“IPO Regulations”) set out several regulatory requirements in relation to IPO by life insurance companies.

Overview

- An insurer needs to obtain a prior written approval from the IRDAI for conducting an IPO. Once the IRDAI approval is received, the insurer needs to approach the Securities and Exchange Board of India (“SEBI”) for its approval for the IPO.
- IRDAI approval has a one year validity within which the insurer would need to file the Draft Red Herring Prospectus (“DRHP”) with the SEBI.

IRDAI Approval – Key Considerations

- The insurer should have completed 10 years of operations before applying to the IRDAI for seeking approval for an IPO. As the Company commenced operations in 2001, the Company meets this criterion.
- IRDAI requires an Embedded Value (“EV”) Report to be prepared by an independent actuarial expert and reviewed by another independent actuary. The EV is required to be calculated in the manner prescribed by the Actuarial Practice Standard issued by the Institute of Actuaries of India (“the IAI”).
- IRDAI expects the EV to be at least 2 times the paid up capital invested in the business. The current EV is in excess of 3.25 times the paid up capital.
- It is noted that the current EV calculations are based on a “Traditional EV” approach whereas the relevant IAI practice standard prescribes a “Market Consistent” approach. It would be unclear how EVs under the two methods compare as there is not sufficient information available. However, the Company definitely needs to re-calculate the EV based on the IAI practice standards.

IRDAI Approval – Other Important Considerations

- In addition to the EV, the IRDAI will also have regard to other considerations such as:
 - Company’s track record in relation to regulatory compliance, particularly in relation to limits on management expenses
 - Compliance with public disclosure requirements and corporate governance guidelines
 - Maintenance of solvency margins in the immediately preceding 6 quarters

- Company's track record in relation to the policyholder protection as evidenced by grievance resolution over the last five years and any regulatory action initiated by the IRDAI.
- IRDAI reserves the right not to accord the approval if, in its opinion:
 - the applicant company is not compliant with the regulatory framework; or
 - where it may be detrimental to the interests of policyholders; or
 - it may not be in the interest of the insurance business in India.
- While according the approval, the IRDAI may prescribe:
 - the extent to which promoters shall dilute their shareholding
 - the maximum subscription which could be allotted to any class of foreign investors
 - minimum lock-in period for the promoters from the date of allotment of shares without prejudice to SEBI requirements in this regard
 - disclosures in the prospectus/offer document in addition to disclosures required by the SEBI
- Annexure 1 to the IPO Regulations sets out detailed disclosure requirements to be incorporated in the DRHP of a life insurance company.
- The regulations state that the IRDAI will grant the approval as expeditiously as possible without specifying an exact timeframe for the approval. [10]

(ii)

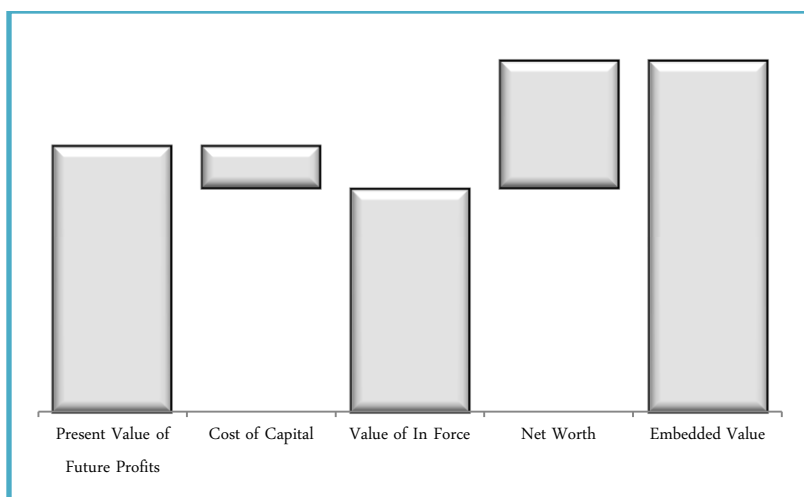
a) Need for calculating the Embedded Value ("EV")

Accounting profits do not appropriately reflect the economic value of business because the Company incurs significant new business strain due to high initial commissions and expenses, whereas profits only emerge gradually over the lifetime of a contract. This is particularly true for unit-linked business which has been the mainstay business line for the Company.

The EV is a more appropriate metric that reflects the underlying economic value of the business.

Components of the EV

The chart below depicts how the Company's EV is currently constructed:



In brief, the EV is the sum of two components; namely Value of In-Force (VIF) and Net Worth. Note that EV reflects the value of business already written by the Company and is still in-force as at the valuation date. The EV does not include value of the new business that the Company is expected to write in the future.

The VIF reflects the present value of future after-tax profits attributed to shareholders from the in-force business, less the cost of holding the capital required to support the in-force business. The future profits are projected using a realistic actuarial assessment¹ of expected future experience in terms of lapses, mortality and expenses. In addition, the projected investment income reflects the Company's target investment strategy and expected return on various asset classes.

It is important to note that the underlying cash flows for insurance business have a fair degree of uncertainty. Some examples of this would include: (a) a pandemic resulting in more deaths than forecast; (b) a sharp rally in the equity markets leading to a "profit-booking" behaviour amongst the unit-linked customers of the Company, which ultimately results in higher surrenders than forecast.

Such uncertainty is allowed for by discounting the projected future after-tax profits attributed to shareholders at a risk discount rate that is higher than the 'risk-free' rate such as G-Sec yields.

Regulators require the Company to hold a certain level of capital, often called Solvency Capital, to support the business. The Cost of Capital reflects the opportunity cost - i.e. the excess of risk discount rate over the after-tax investment income earned on the Solvency Capital – associated with holding this capital.

Finally, Net Worth is simply the total assets that the Company has after deducting the assets that are earmarked to support the liabilities. In calculating the Net Worth, certain adjustments may be made to reflect the underlying economic value of assets:

- Intangible assets such as ‘goodwill’ (if any) may be excluded.
- Market Value Adjustments may be incorporated for assets that are not held at the market value in the balance sheet.

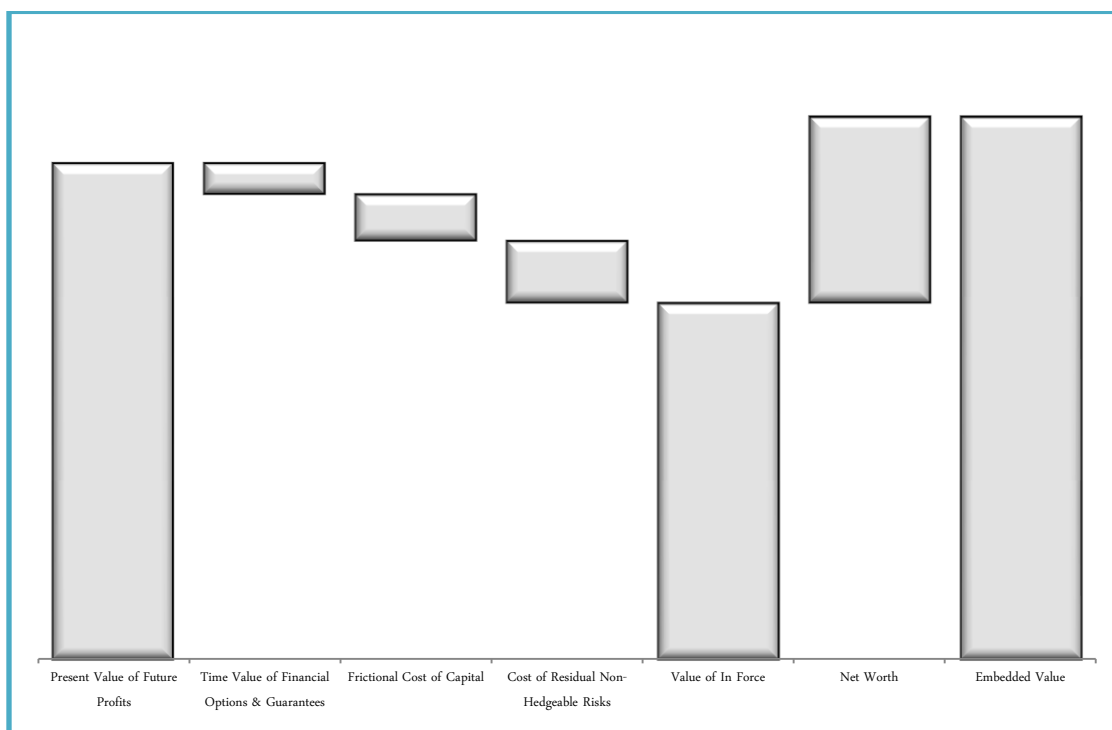
[5]

- b) IRDAI’s IPO Regulations state that the insurer’s Embedded Value should be calculated in the manner prescribed by the relevant Actuarial Practice Standard (“APS”) issued by the Institute of Actuaries of India (“IAI”). The relevant APS for this purpose is the APS-10.

Methodology prescribed under APS-10

The methodology prescribed under APS-10 is termed as ‘Indian Embedded Value’ (IEV).

The chart below depicts how the IEV is constructed:



All the above components are required to be disclosed separately.

Company's Current TEV Methodology vis-à-vis IEV Methodology

The table below summarises the key differences between the Company's current TEV methodology and the IEV methodology under APS-10. A brief commentary follows the table.

Particulars	Current EV Approach (TEV)	IEV Approach
Independence	Current approach is to develop the TEV results internally within the Company	APS10 requires two independent actuaries ('Reporting Actuary' to develop and 'Reviewing Actuary' to review) to be involved
Materiality	No explicit requirements for the TEV calculations to conform to any stated benchmarks	The Board of Directors are expected to specify the materiality limits (as a % of IEV) and the calculated EV should be within the materiality limit specified.
Economic Assumptions	- Based on target asset mix and expected long term returns on various asset classes - Allows credit for equity risk premiums for setting expected returns	- Based on a 'market consistent' approach - Expected investment returns reflect 'risk-free' returns - No credit for equity risk premiums
Allowance for risks	Risk discount rate which is typically higher than the 'risk-free' rate	The Present Value of Future Profits is computed using the 'risk-free' rates. However, the risk margin is allowed through an explicit deduction termed as "Cost of Residual Non-Hedgeable Risks".
Cost of Guarantees	- No prescribed standard - Can be determined using sophisticated stochastic approaches or approximated using simplified approaches (e.g. as a % charge of reserves or % charge of premiums) - Not required to be	- Stochastic approaches prescribed - Explicit disclosure required

	disclosed separately	
Cost of Capital	Based on calculation of the opportunity cost of holding the capital by taking into account excess of risk discount rate over after-tax investment income on the solvency capital	Only reflects frictional costs i.e. investment expenses and taxes on investment income associated with holding the solvency capital
Expense forecasts	No prescribed standard.	Expense assumptions should be based on current level of expenses. Future productivity improvements cannot be allowed for in the projections.
Disclosures	No explicit disclosure requirements	Detailed disclosure requirements, including minimum sensitivity analysis and analysis of movement of embedded value over the period are specified in APS 10

In brief, the IEV methodology is based on ‘market-consistent’ principles wherein risk quantification is explicit as opposed to the risk margins being implicit within the risk discount rate under the TEV approach. As a result, the IEV assigns greater importance to the inherent risk characteristics within various product lines than the TEV. **[10]**

(iii) Reasons for the declining margins of unit linked business:

- IRDAI has set an upper limit on the charges that can be levied on unit-linked products, often referred to as “Cap on Charges”. The Cap on Charges was initially introduced in January 2010 and further strengthened in September 2010 amongst other things by extending the same to cover the surrender penalties.
- Although the Cap on Charges is not unreasonable as far as maturing policies are concerned, there are two constraining limits.
- Firstly, the Company can only set charges equivalent to 4% p.a. for the initial five years. In other words, if the underlying unit-linked funds earn 10% p.a. for the first five years, the client needs to receive a return of at least 6% p.a.. Moreover, the pattern of charge deduction needs to be reasonably even (although not necessarily equal) during the five years. Note that this results in a mismatch between the charges received and the actual expenses incurred (which are high at the point of sale).

- Secondly, the limit on surrender charges is rather stringent. The surrender charges are limited to levels fairly below the initial expenses that the Company incurs in writing new business, particularly for the Agency channel. As a result, the Company is far more exposed to lapse risk than before. Typically, a unit-linked policy needs to remain in force for five to six years for the Company to break-even. However, managing good persistency has proven to be a very stiff challenge.
- In response to the Cap on Charges, the Company has restructured the products significantly and the levels of commissions have reduced. That said, for channels such as Agency, where the costs are not fully variable, the expected level of charges after taking into account a realistic assessment of persistency experience is barely sufficient (often insufficient) to finance the initial commissions and expenses leading to erosion of margins. The productivity levels of Agency channels would need to be far more for the profitability to return anywhere near the previous levels.
- Over the years, the persistency experience has only worsened. This is attributable to a variety of reasons: distribution mis-selling in the previous equity bull market, short term outlook of the customers particularly in view of outperformance of asset classes such as property and gold and of course the fact that regulatory surrender charges are too low to be an effective exit barrier. VNB margins have come down as this deterioration started getting reflected in the VNB basis over time.

Steps that the Company can take to improve the VNB margins for unit-linked business

- Perform a detailed analysis on VNB margins for unit-linked business across various channels. It may be the case that the VNB margins for relatively inefficient (in terms of expenses vis-à-vis revenues) channels are low or negative. In such circumstances, it may be a good idea to manage the channels such a way that the inefficient channels are discouraged to market unit-linked products and instead sell traditional products. Overall, this would improve the VNB margins for unit-linked business as well as that at the Company level.
- Consider whether commission arrangements with bank distributors can be re-negotiated to pay lower initial commission and in return pay higher renewal commissions (within the regulatory caps). Such an arrangement would reduce the lapse risk and provide uplift to the VNB margins.
- Another option that achieves a similar result could be to evaluate the initial commission claw-back rules in case of early lapses / surrenders.
- As a result of regulatory limits on surrender charges, persistency is a critical driver for VNB margins. The Company can undertake measures to improve persistency through:
 - Further strengthening the quality of disclosures at the point of sale
 - Ensuring that the market conduct of the distributor at the point of sale is appropriate. At times, distributors sell unit-linked products on the basis of high returns in the recent past. When the actual returns don't match up to the

high returns discussed at the point of sale (even though actual returns are reasonable), the disillusion often triggers surrender.

- Establishing a separate retention/persistency team that focuses on improving persistency
- Focus on regular pay unit-linked products as opposed to short pays which typically have far lower VNB margins
- Evaluate the elements within the product design that encourages long-term persistency such as loyalty additions. Existing policyholders may be reminded of such benefits periodically (e.g. along with the annual account statements) to encourage them to remain invested.
- Increase the protection element in the product through optional health and accident benefits on top of the life cover embedded in a unit-linked product. Protection covers typically have a far higher VNB margin than savings component of a unit-linked product and this can, therefore, provide an uplift to the overall VNB. Commissions may also be partially restructured to drive distribution behaviour towards selling more protection elements.
- Remain focussed on the performance of the underlying funds vis-à-vis corresponding benchmarks. Demonstration of solid fund performance can be a powerful tool to retain customers.
- Remain focussed on expenses in general and keep these under control
- Market unit-linked products directly to the customers through online channel resulting in saving of distribution commissions and potentially enhancing the levels of persistency. Both these may help increase the VNB margins. [10]

(iv)

a) The biggest challenge is the mismatch between the assets and liabilities emerging from the fact that the liabilities i.e. customer benefits are defined at the outset, whereas the asset returns cannot be predicted with certainty as these would depend on reinvestment yields as and when future premiums are received.

The Company's exposure to the ALM risk is the greatest at the start of a policy, as most premiums are yet to be received and invested. With the passage of time, as premiums are received and invested, the exposure to ALM risk would reduce. After all premiums are received, the Company could deploy duration matching or cash flow matching techniques to facilitate robust ALM for the block.

As the Company's exposure to the ALM risk is higher in the initial years, the business can be quite vulnerable to a fall in yields. The assets and liabilities are not matched and therefore a fall in yields could trigger an increase in reserves disproportionately higher than the increase in the value of fixed income assets (assuming such assets are realised, however, in which case there will also be a re-investment risk). As a result, this could introduce volatility in accounting income as well as the solvency position of the Company depending on the volumes of such product relative to the overall balance sheet.

On the other hand, if interest rates increase significantly in the early years, this could encourage policyholders to discontinue premium payments and invest the money in other high yielding products. This could lead to lapse experience being adverse and loss of future profitability.

The average premium paying term is 12 years, although the Company does offer shorter premium paying terms. Arguably, this reflects distribution behaviour as the initial commissions are optimal at 12 year premium paying term. However, the resulting ALM risk is far higher than, if the short payment products were written more.

The Company has only recently started writing this business and the share is relatively small so this helps. However, the share of this product is increasing over the last few years.

Earnings and solvency volatility may not be desirable from the standpoint of a Company thinking about an IPO.

IRDAI has recently allowed insurance companies to enter into fixed income derivatives for the purpose of hedging. Admittedly, this helps mitigate reinvestment risk but not before introducing other issues:

- Often, these contracts are Over-The-Counter (OTC) with bankers. In addition to the hedging cost, the Company would also be exposed to counter-party risk.
 - Structuring of such contracts is based on cash flows locked down at the outset. This exposes the Company to the risk of actual lapses turning out to be significantly different than estimated in the construction of the hedge.
- [6]

b) Equity returns are volatile, but are expected to outperform fixed income instruments in the long run. A high level analysis of historical data suggests that viewed over a 10 to 15 year horizon, Indian equity indices quite often outperform fixed income returns. Prima facie, therefore, the suggestion is not completely unfounded. That said, there are several issues that the Company needs to consider for the purpose of setting an appropriate investment strategy.

Even over a longer horizon, equity outperformance is not guaranteed.

Under the non-participating savings products, the policyholder returns are fixed at the outset. Investing a part of the reserves into equities essentially makes this a leveraged play – borrowing money from policyholders at a particular fixed rate and investing it into equities, with policyholders having no participation in the upside as well as downside. As is typical of the leveraged investments, these could lead to significant gains or losses

depending on the equity performance. However, the shareholders and regulators typically expect companies to make *spread profits* and not leverage profits.

Although the average policy term of the underlying business may be on the longer side, the actual duration may not be that long. A realistic assessment of the expected discontinuance experience may be particularly useful in this regard. For instance, if a realistic assessment is that 50%-60% of the policyholders exit within a period of five years from the policy issue, then the liabilities would become rather short term to merit any consideration of equity backing.

Most importantly, such products inherently have a mismatched asset liability position due to the reinvestment risk associated with future premiums. An investment strategy involving equities further worsens the asset liability mismatch. Quite obviously, such a mismatch brings several issues to the fore:

- Earnings volatility from one year to another depending on equity performance (realised gains) during the year
- Solvency ratio volatility that would stem from earnings volatility
- Need for setting up asset-liability mismatch reserves which could deplete the profit margins
- Potential for increased regulatory scrutiny particularly in adverse economic environment. Quite often, an equity market crash is correlated with a fall in yields as people scamper to exit equities towards the safety of G-Secs. This could be a double whammy for the Company to handle if it happens in the initial years.

Overall, such an investment strategy is normally not desirable and particularly so for a Company that is thinking about an IPO. [9]

[50 Marks]

Solution 2:

i) Policyholders' reasonable expectations (PRE)

This relates to policyholders' reasonable expectations with regard to the level of benefits or charges under contracts where these are at the discretion of the life insurance company. There is no generally accepted definition of PRE, but they will be influenced by, for example, the past practice of a company and any literature issued by it, particularly the PPFM. Judgement will be called for in managing PRE. The concept of PRE is closely linked to Treating Customers Fairly (TCF).

- a)** PRE for participating products may be measured and managed by:

- Level of benefits and bonuses that may be set out in product brochures, examples of policyholder benefits that may be included in marketing materials and benefit illustrations provided at the point of sale.
- Company's communications regarding overall returns that may be earned on the products.
- Marketing campaigns and general press articles comparing available bonuses / overall returns with other companies and alternate forms of investments.
- Company's marketing and other literature will also set out how it intends to distribute surplus.
- Communications with respect to how the Company intends to split different types of bonuses and how bonuses shall be determined.
- Company's own past practice
- Practices of other companies in the industry
- These may be managed by greater emphasis on attempting to pay 100% or close to 100% of asset share on surrender and maturity.
- Further manage by consistent distribution of surplus between generations of policyholders by operating a documented philosophy of smoothing

b) Most of the benefits in these products are expected to be contractual and guaranteed at outset. The PRE, therefore, applies to the remaining aspects of the products or benefits that are not guaranteed / discretionary. These can be measured / managed by:

- Marketing literature, point of sale illustrations and communications, past practice and common market practice of competitors apply for non-participating products as much as for participating products mentioned above.
- Other specific areas of measuring and managing PRE for non-participating products might be:
 - Ensure smooth claims process for promised benefits upon contingent events, particularly for protection products (as most insurance marketing campaigns emphasis on “peace of mind” – so it would be reasonable for policyholders to expect efficient claims procedures that such “peace of mind” is maintained at times of distress, which should be managed by establishing seamless operational procedures)
 - Measure PRE in respect of overall financial stability of the company to meet its guaranteed obligations based on public disclosures of company's financial health and operating experience. Manage the same by establishing appropriate risk management practices and sufficient provisioning to ensure that the company is able to deliver on its promises.
 - Clear illustrations and communications of any discontinuance benefits that might be available during the policy term.

c) Most of the charges in these products are expected to be contractual and guaranteed at outset. The PRE, therefore, applies to the remaining aspects of the products and benefits that are not guaranteed / discretionary. These can be measured / managed by:

- Marketing literature, point of sale communications, past practice and common market practice of competitors apply for unit-linked products as much as for participating.
- Other specific areas of measuring and managing PRE for UL might be:
 - Comparison of overall difference between gross earned yield (before deduction of all charges) and net return to the policyholder (after deduction of all charges). The overall difference should be reasonable and comparable with overall charges on alternative investments
 - Clear point of sale communication in respect of absolute quantum of charges applied and any restricting future revisions to the charges during the policy terms. If revisions are considered necessary, accompany these with appropriate communications and set up operational mechanisms to address customer queries
 - Comparison of overall returns earned on the unit-linked portfolio with the capital market indices for chosen class of assets. For example, if a policyholder chooses to invest in “growth” funds that invest in equities, then his returns would reasonably be expected to mirror returns on benchmark indices
 - This may be managed by strictly adhering to communicated mandate for each fund and rigorously evaluating returns against stated benchmarks
 - Ensuring appropriate returns on invested premiums in case of surrenders. Manage by suitable communications of applicable surrender charges as well as emphasis on minimising (or eliminating) such charges after a reasonable length of initial period
 - Ensuring that there are no errors in the unit-pricing processes and systems and any issues are identified and corrected immediately upon discovery.

[9]

(ii)

(a) The following checks and validations may be undertaken:

Validating asset share methodology:

- Check completeness of coverage of asset share methodology: have all relevant cash-flows been allowed for in the accumulation of historic asset shares? In particular, validate whether accumulation taken into account all product features and allowance is made for deductions for expenses, commissions, cost of

providing life and other contingent cover, cost of providing any guarantees or options on the product, tax (for life business only), cost of capital etc.

- Have appropriate contribution for special conditions been allowed for? For example, contribution to asset share from surrenders, paid-up's, other permissible benefit alterations etc.
- Have appropriate contribution to asset shares been allowed for from other sources – e.g. riders?
- Check whether allocation of expenses to asset shares is reasonable relative to total expenses of the company and the expense allocated to the participating fund.
- Check whether the asset share methodology contravenes any tenets of TCF. In particular, check that the approach adopted in calculating the asset shares for the purpose of setting the bonus rates at the point of sale is consistent with the approach adopted in recent calculations.
- Check application of tax is consistent with the applicable income tax rate for the relevant years.
- Check appropriateness of granularity of the asset share calculation – are the asset shares calculated for sample representative model points / policies or are these calculated individually for all policies since inception? In case calculated on representative model point basis, validate whether the chosen sample data provides sufficient representation for the underlying population and whether all features are covered.
- Validate allowance for shareholder transfers in the asset share methodology. Have part of historical surpluses been transferred to the shareholders – in which case, these need to be appropriately deducted.
- Validate methodology applied against other company documentation – for example, this should be consistent with the Company's PPFM – if established.

Validating assumptions:

- Most important assumption is likely to be the choice of accumulation rate. Validate this against the actual rate of return earned on investment, for each year individually. Depending on the purpose, the return should be on a book value basis (ignoring unrealised gains/losses) or market value basis (allowing for unrealised gains/losses) – to validate the same against other documents.
- As an additional sense check, compare the earned rate with the underlying asset mix and de-compose the overall rate by return earned on asset types. Validate independently whether these are reasonable relative to capital market benchmarks. It is possible that the individual returns have out- or under-performed relative to benchmarks, but this is a helpful reasonableness check nonetheless.
- The other important assumption is likely to be the amount of expenses loaded for each historic year. As part of methodology checks, should already validate the

allocation between participating and non-participating lines of business. In addition to this, check the level of expenses loaded in the asset share methodology against those used in setting out benefit illustrations and initial pricing. If these are materially out of line, then consider whether this is in line with PRE and TCF.

- Assumptions in respect of commissions should reflect actual commission payments made. Check these against commission rates set out in product filings. In addition, check whether there are any commission savings from non-payment of commissions to distributors due to termination of the relevant distributor. Allow for this in the overall commission rate established for the asset share calculation.
- Check mortality decrement applied against actual experience of the company. Check granularity of the experience analysis used in determining the assumptions for the asset shares – are the assumptions based on overall line of business experience or product-wise experience?
- Are there any applicable cohort effects that should be applied? Ensure that granularity of mortality assumption applied for the asset share calculation is consistent with that used at the time of pricing the contract. For example, if varying premium rates were charged for the cost of death based on gender or other underwriting variables (such as smoker status), then it needs to be ensured that the cost of risk applied on the asset shares is consistent with this.
- Check treatment of sub-standard lives in deriving the mortality assumption. If these have been charged extra premium, then these may need to be allowed for separately.
- Check assumptions for other decrements such as surrenders and paid-up's (if relevant) against actual experience.

Validating calculations and results:

- Set up independent sample model point checks to independently calculate the accumulated asset shares with the chosen methodology and assumption. This may be done for a small number of model points only – to flush out any possible mathematical or interpretational errors in the company's calculations.
- For aggregate asset shares of the company, as a first check – validate the completeness of the data used. This may be undertaken for key summary statistics to ensure that all relevant data has been captured. Compare the summary statistics underlying the results (example number of policies, premium, benefit amounts etc.) against independent data records maintained within the company.
- As a corollary to the above, validate whether all relevant products have been captured and no product class has been inadvertently missed in the calculations.
- Graphically chart the development of asset share over time to visually validate the development over time. This would highlight the timing of the particular cash-

- flows based on product features and help validate that such have been appropriately reflected.
- Check reasonableness and internal consistency of results by comparing asset shares calculated for cohorts of policies and noticing how these compare relative to each other.
 - Check overall reasonableness of calculated asset shares relative to total assets available within the participating fund.
 - Check overall reasonableness of calculated asset shares relative to the GPV reserves for the corresponding class of business.
 - Check overall reasonableness of calculated asset share by comparing with accumulation of the emerging (revenue account) surplus within the participating fund for each year.

Compliance:

To review compliance against the following key aspects:

- The IRDA (Non-Linked Insurance Products) Regulations, 2013 require that:
 - Asset shares are maintained at a policy level
 - Only the portion of expenses representing this business shall be allocated to and interest rate credits to these asset shares shall represent the underlying assets of these funds.
 - The Appointed Actuary shall be responsible to determine the asset share for each product in accordance with the guidance or practice standards etc. issued by the Institute of Actuaries of India.
 - The detailed working of the asset share, the expensed allowed for, the investment income earned on the fund etc. which are represented in the asset share shall be approved by a with profits committee.
- Guidance Note 6 of the IAI, Management of participating life insurance business with reference to distribution of surplus, sets out adoption of asset shares as recommended practice for actuaries involved in life insurance work. Specific requirements of the guidance note are as follows:
 - Asset shares, where calculated, should be separately determined for each policy grouping.
 - The policy grouping will be influenced by any risk sharing rules that the insurer has, either explicit or implicit. For example, if the risk sharing rules dictate that the investment experience will be shared differently between single premium and regular premium policies, the asset share will be determined separately for single and regular premium policies.

- The GN requires the Appointed Actuary to document the company's approach to and treatment of each of these items of surplus or deficit for the purposes of deriving the asset shares. In particular the Appointed Actuary should consider whether the surplus or deficit item forms part of the asset shares or is maintained within the fund's estate.
 - Consider historical experience. Only deviate if sufficient information is unavailable.
 - If allowing for surrenders in the asset shares, this may be explicit or implicit (e.g. through adjusting the investment returns)
 - The method of calculation of asset shares should be fair and appropriate and consistent from year to year
 - In setting assumptions, wherever possible, the Appointed Actuary should make use of actual historical data and cash flows to derive the historical asset share
 - For the purpose of asset share calculations, shareholders' transfers refer to the shareholders' share of the cost of bonus. This can be attributed to the relevant product grouping to determine the asset share for each policy grouping. Asset shares would not typically reflect any transfers into the fund to support new business strain etc.
 - The acquisition expense overruns should not be charged to the asset shares. The allowance for maintenance expense overruns can be made provided PRE encompasses expenses risk.
- The Appointed Actuary should keep abreast of changes in tax rules, and ensure that derivation of asset shares are consistent with the appropriate tax rules prevailing.

[12]

(b) The implications of writing non-participating business with significant capital strain and using the par fund FFA to support such a strain on the various performance indicators used by the Company are discussed below:

i. annual growth in new business volumes

- The primary driver for new business growth is unlikely to be whether the product is written within the participating fund or the non-participating fund, but rather more linked with the attractiveness of the specific features of the product and distribution capabilities. Insofar as this is the case, choice of source of funding of the capital strain should have a limited impact on the growth in new business volumes.

- Nevertheless, the proposed new product is capital intensive and the Company has accumulated a large estate. If the ability of the Company to write new business is constrained by availability of sufficient capital to support the new business strain, then writing the new product within the participating fund may enhance the ability of the Company to sell more volumes.
- However, to the extent the par fund FFA is used to support the non-par business' capital strain, the ability of the Company to write significant volumes of new business in the participating fund itself may be constrained.
- The impact on new business growth of writing the product in the par fund is only second order and depends more on the degree of capital constraint within the Company.
- Even if the product is not written within the participating fund, some credit may be available from the par fund FFA to support the solvency margin requirements of the non-participating business. For instance the current regulations in India permit use of assets within the participating fund for demonstrating overall solvency of the Company and backing the solvency margin requirements of the non-participating products. Thus, part of the capital constraint may be lifted by using par fund FFA for backing the non-par fund solvency margin requirements – even if the product is not written in the par fund. However, this does not extend to supporting reserving requirements, so the advantage of writing the product in the par fund from the perspective of new business growth would again depend on the overall level of capital constraint.
- In terms of overall growth of new business, if the Company expects high profit margins from the new product within the par fund, then in addition to the sale of the new non-par product, it may use the decision to write the high margin product in the par fund to boost sales from the par business as well. This would only be achievable if the Company is able to convince the potential new par policyholders that investing in the par products would provide them the additional benefit of sharing in the surplus from the profit margins of the non-participating products too. However, this is a complicated strategy and would require sophisticated sales staff as well as customers and may also lead to mixed messages about relative attractiveness of different product propositions.

ii. bonus rates declared

- It is noted that the Company already has a large estate (or FFA) in the participating fund. So in the short run, insofar as the Company has been declaring supportable bonuses historically and continues to declare smoothed bonuses, these should continue as per established PRE as the current FFA may not be required to be tapped on to support future bonuses other than for smoothing purposes.
- However, if the Company intends to increase the bonus rates in the near short term to distribute some of the FFA through higher bonuses, then it would not be

able to do so to the extent envisaged, if the FFA is used to support the capital strain from the new non-participating product.

- If the NB strain from the non-participating product is so significant (due to volumes sold) that it dwarfs the current FFA (even though it is stated that it is already large), then there is a possibility that the available FFA within the participating fund gets absorbed in supporting the new product and restricts the Company's ability to declare on going bonuses in the short run.
- In the long run, if the high margins on the non-par product materialize, then it would improve the overall FFA available in the participating fund and increase the Company's ability to declare higher bonuses to the participating products.
- The bonuses in the long run also depend on the investment strategy for the FFA. If these are invested in high yielding assets, then there is a risk of loss to the fund from the possible high returns, if the FFA is used towards supporting the non-participating business. However, if the current asset allocation is relatively conservative, this would not be very relevant.
- There is an additional risk – if the expected margins on the non-participating product do not materialize due to adverse future experience. In this case, the ability of the Company to smooth the bonuses will be restricted as the loss shall be borne by the par fund FFA.

iii. IRDAI defined solvency ratio

- This should remain unaffected regardless of whether the non-participating product is written in the par fund or the non-par fund.
- This is because as per the IRDA defined solvency margin rules, the solvency of the Company is measured at the Company level by taking into account the total assets (including the FFA). Thus, the choice of fund in which the product is written should not impact the overall ratio.

iv. amount of statutory profits for the year

- The total statutory profits are a function of multiple components and would depend on the interplay of each one of these.
- Other things remaining unchanged, if writing the non-participating product in the par fund results in greater new business volumes (by alleviating the capital constraint discussed above), then initially this should result in significant NB strain, reducing the statutory profit for the year in the immediate short term.
- However, high margins on the product would be expected to emerge over time (on the assumption that the product is priced profitably and the actual experience bears out), so it would be expected that the statutory surplus emerging in the long run would increase – however this would be within the par fund.

- If written in the par fund, as the margins emerge from the product, depending on how PRE is set, it may be a reasonable expectation of the participating policyholders that they benefit from this profit through higher bonuses. Such bonuses would imply greater distribution of emerging surplus for the participating policyholders (as their reward for initially supporting the capital requirements of the non-participating product) and consequently from the Company's perspective, it would result in lower corporate profits since now a greater amount of benefit is required to be paid to the par policyholders.
- By contrast, if the product was written in the non-par fund, in the long run the statutory profits for the Company would be higher (again assuming the product is priced profitably and the experience is borne out) as the margins from the product emerge.

v. EV (calculated following TEV methodology)

There are two critical components of EV that we need to consider:

- First, the impact of writing the new non-participating product in the participating fund on distributable shareholder earnings. When writing high margin products, in general, it would be expected that the EV of the Company would increase over time. However, if this product is written within the par fund, only 10% of the emerging surplus would be attributable to the shareholders, so the EV calculated (as the discounted value of shareholder profits) would not increase as much as it would have, had the product been written in the non-participating fund.
- At the same time, we need to consider whether the Company would be able to write the new product at all, if this is not written in the par fund (due to capital and other constraints). If it is the case, that the Company is unable to write the product unless written in the par fund, then even 10% emerging surplus would be greater than what would otherwise be achieved and from that perspective results in a better shareholder outcome.
- Second, the impact of writing the new product in the participating fund has on the cost of capital. As noted above, the Company may utilize FFA to support solvency margin of all business (and not just par) – so a high FFA may contribute to reducing the cost of capital to an extent. However, the degree to which this is the case depends on internal governance principles of the Company, which may restrict the use of participating fund FFA to support other lines of business, unless explicitly written within the participating fund. If this is the case (i.e. there is indeed an internal restriction stating that the FFA can support other lines of business only if they are written within the participating fund), then again writing the new business within the participating fund may would help reduce the cost of capital. This may, of course, be limited due the fact that the level of FFA itself

may come down if it is used to support the new business strain of the non-participating product. The net impact on the EV of the Company may be positive or negative. However, if this is not the case (i.e. there is no internal restriction stating that the FFA can support other lines of business only if they are written within the participating fund), then there is unlikely to be a CoC impact based on the choice of fund used to write the new product in.

[16]

c)

- The IRDA (Non-Linked Insurance Products) Regulations, 2013 has noted that each Company shall constitute a with-profits committee ('WPC') for the governance and management of the with-profit fund.
- As a minimum, the regulation requires that the WPC consists of one independent director of the Board, the CEO, the Appointed Actuary and an independent actuary.
- The regulation further requires that the report of the WPC shall be appended to the Actuarial Report and Abstract that is submitted to the regulator annually by the Appointed Actuary.
- The regulation requires that the WPC approves the detailed working of the asset share, the expenses allowed for, the investment income earned on the fund etc. which are represented in the asset share.
- Outside of this, the regulations are largely silent with respect to the detailed scope, and other roles and responsibilities for the WPC. Consequently, companies have applied varied interpretations as to the precise (extended) scope of the WPC.
- Some Companies have restricted the role of the WPC purely to a sign off of the Company's asset share.
- However, best practice – in the spirit rather than letter of the regulations – in this regard indicates a wider mandate for the WPC, which should be involved in overall governance of the participating fund. Such a 'extended' scope may include:
 - Interpretation of PRE and formalizing the Company's approach to demonstrating that it meets the PRE
 - Instating robust governance principles for management of the with-profits business, including approach to determining and declaring bonuses, establishing a bonus philosophy covering distribution of surplus, establishing a smoothing policy etc.
 - Overseeing the investment policy of the WP fund

- Establishing principles to determine target benefits in respect of surrender and maturity values
- Investigating financial condition of the with-profits fund, including ascertaining sufficiency of reserves, adopted risk management principles etc.
- Reviewing the documentation of such principles and practices adopted.

[6]

(d)

The PPFM is a governance document that is intended to provide a governance framework for the company's with-profit business as well as provide protection to the policyholders from any unfair practice by the Company.

Principles are enduring statements of the overarching standards that a company adopts in managing its with-profits funds and so should be subject to very little change.

Practices however, must contain enough detail to enable a knowledgeable observer to understand the material risks and rewards from effecting or maintaining a with-profits policy with the Company.

So the 'Principles' describe the approach that may be used by the Company in responding to longer-term changes in the business and economic environment, whereas the 'Practices' describe the approach that may be used by the Company in responding to shorter-term changes in the business and economic environment. Hence, the 'Practices' may change more frequently than the 'Principles'.

The PPFM would typically set out how the with-profits business is conducted in each of the following areas:

- **Determining the payouts to policyholders**

This covers the methods and approximations used, documentation of the process, and procedures for making changes to methods or assumptions. This would include:

- Annual bonus policy
- Terminal bonus policy
- Smoothing policy

- **Investment strategy**

This covers the degree of matching, the approach to be adopted in investments of different currency, liquidity and volatility, and controls on using new investment instruments.

- **Business risk**
This covers how the Company decides on whether a particular business risk is acceptable, arrangements for limiting the scale of such risks, and how the resulting profit or loss is reflected in payouts on with-profits policies.
- **Charges and expenses**
This covers how charges are applied and how expenses are apportioned to with-profits business.
- **Management of the inherited estate**
This includes how the inherited estate may be used.
- **Volumes and type of new business**
For open funds, this covers how limits on the quantity and type of new with-profits business are reviewed.
- **Closure to new business**
This covers actions the company would take if it closed to new business.
- **Achieving equity between policyholders and shareholders.**
This includes a description of any profit-sharing arrangements. It must also cover any other areas that are considered important to the management of the Company's with-profits fund(s) and that may affect payouts to with-profits policyholders.

[7]

[50 Marks]
